NIAGARA 4 · API COMPARISON

Niagara N4 API Comparison

oBIX · **Gline REST API v1 + v2** · BTIB ACTIVE-API · Tridium /rest/ · Third-Party Analytics

Side-by-side comparison of approaches to interact with Niagara N4 stations.
All Gline REST figures verified live on a Niagara 4.14 station.

Shanghai Gline Net Co. · 2026

FEATURE MATRIX

Gline REST API — v1 vs v2 vs the alternatives

Feature	oBIX (built-in)	Gline REST v1 (free, this repo)	Gline REST v2 (on request)	BTIB ACTIVE-API	Tridium /rest/	Analytics
Read values	Yes	Yes	Yes	Yes	Yes	Yes
Write values	Yes	Yes	Yes	Yes	Yes	No
Create / delete components	No	Yes	Yes	Limited	Limited	No
Deep copy (link rebuild)	No	Yes	Yes	No	No	No
Rename (link-safe)	No	Yes	Yes	No	No	No
Link / unlink	Partial	Yes	Yes	Yes	No	No
Invoke actions	POST to action	Yes	Yes	Yes	No	No
Atomic batch (single POST)	No	11 ops	12 ops	Limited	No	No
BQL query → JSON v2	No	No	Yes	No	No	No
Program management v2	No	No	Yes	No	No	No
Built-in auth (Basic) v2	SCRAM / Basic	No	Yes	Basic / Token	Basic	API key
CIDR allow-list v2	No	No	Yes	No	No	No
Alarm / history query	Yes	No	No	Yes	Limited	Yes
License cost	Free	Free	Free*	$462–$1k	SMA	SaaS
SMA required	No	No	No	No	Yes	Yes
Port	80/443	8081	8082	80/443	443	SaaS

* v2 is distributed on request. Both Gline editions are independent modules and can run side by side.

GLINE REST API v2

Gline REST API v2 — what it adds NEW in v2

v2 = v1 + authentication + access control + BQL + program management.

New in v2	What it does
BQL query → JSON	Run BQL over HTTP and get a structured table: `{"columns":[...],"rows":[...],"count":N}`. Ready for reports, dashboards and AI pipelines — no XML parsing.
Program management	`POST /api/program/write`, `POST /api/program/addslot`, `GET /api/program/status` — write source, add slots, read status over HTTP.
Built-in authentication	HTTP Basic Auth, configurable at runtime. Missing or wrong credentials → **401**.
CIDR allow-list	Restrict access to trusted network segments (comma-separated ranges). Outside the list → **403**. Loopback `127.0.0.1` always allowed as a recovery path.
Persistent config	Auth and allow-list configuration survives station restarts. No external reverse proxy needed.

Verified on v2.0.3, Niagara 4.14. **40 / 40 test cases passed** (29 functional + 11 access-control).

ENDPOINTS

Gline REST API — endpoints

Endpoint	Method	Edition	Purpose
`/api/status`	GET	v1 · v2	Service status (auth-exempt liveness probe)
`/api/modules`	GET	v1 · v2	Module information
`/api/types`	GET	v1 · v2	Component types
`/api/config`	GET	v1 · v2	Browse the station tree
`/api/batch`	POST	v1 · v2	Batch component operations + BQL
`/api/program/write`	POST	v2	Write program source
`/api/program/addslot`	POST	v2	Add a program slot
`/api/program/status`	GET	v2	Read program status

Batch operations (single `POST /api/batch`)

v1 (11): ADD · LINK · RENAME · DUPLICATE · CP · MOD · UNLINK · REMOVE · READ · LIST · INVOKE

v2 (12): the 11 above + **BQL**

Example request (v2)

```
curl -u "<USER>:<PASS>" -X POST http://<HOST>:8082/api/batch -d '[{"OP":"bql","Q":"select * from control:NumericWritable"}]'
```

DECISION GUIDE

Which to use

Gline REST API v2

- You need component lifecycle **and** authentication/access control in one module
- You want BQL results as ready-to-use JSON tables
- You manage programs over HTTP
- You want 11 batch ops + BQL in one atomic request

Gline REST API v1

- You only need the 11 batch operations, no auth layer
- Single `.jar` drop-in, no SMA

oBIX

- Alarm and history extraction (built-in, free)
- You cannot install custom modules on the station

BTIB / Tridium /rest/ / Analytics

- BTIB — budget covers the license; you need commercial support
- Tridium /rest/ — station has SMA; simple JSON read/write
- Analytics — dashboards and reporting, not station management

CONTACT

Shanghai Gline Net Co.

For commercial customization, integration, or deployment support:

✉️ **jason.zhang@gline-net.com**

📱 WhatsApp **+86 138 0199 0968**

All Gline REST figures verified live on a Niagara 4.14 station. Features, availability and pricing of third-party products may vary — verify directly with each vendor.